

Problem-Solution Fit

Date	13 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] Banks and financial institutions issuing credit cards; credit/loan officers who screen applications; fintech lenders looking for a low-cost initial screening tool.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Smaller lenders often lack budget for enterprise credit-scoring platforms (e.g., FICO) or a dedicated data-science team; need a lightweight tool that runs on a standard web server/browser, not specialized hardware.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual underwriting: uses human judgment but is slow and inconsistent. Rule-based approval systems: simple and transparent but rigid, with high false-rejection rates. Commercial credit-scoring services (e.g., FICO): industry-trusted but expensive and largely a black box.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Manually reviewing each applicant's income, employment, and financial history is slow and inconsistent. This happens for every single application submitted (high frequency - potentially hundreds per day for a mid-size lender).	9. PROBLEM ROOT / CAUSE [RC] Manual or fixed-rule assessment of many applicant attributes does not scale and cannot learn from historical repayment/default patterns, leading to inconsistent decisions and slow turnaround.	7. BEHAVIOR + ITS INTENSITY [BE] Officers manually cross-check each applicant's details (income, employment length, family status, etc.) against internal guidelines, spending significant time per case; intensity is high given daily application volume.
3. TRIGGERS TO ACT [TR] A growing backlog of pending applications, rising default rates from inconsistent risk assessment, or applicant complaints about slow approval turnaround.	10. YOUR SOLUTION [SL] A machine learning web app (Flask + Random Forest classifier trained on historical application and credit-history data) that takes an applicant's key	8. CHANNELS OF BEHAVIOR [CH] ONLINE Internal loan-management software, spreadsheets, or email submission of application forms.

4. EMOTIONS

BEFORE / AFTER [EM]

Before: frustration and stress for officers handling large application volumes; anxiety for applicants waiting on a decision.
After: confidence and relief from faster, consistent, data-driven decisions.

details as form input and instantly returns an Approved/Rejected prediction, giving officers a fast, consistent, data-driven decision-support tool.

OFFLINE

In-person branch review, paper application forms, phone-based verification calls.